

How a magnetic flux tube changes across neighbouring sections

Three tensor parts. Each column isolates one positive coefficient.

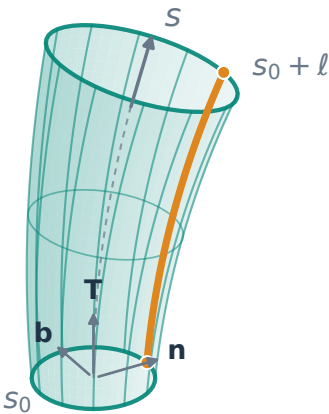
$\mathbf{T} = \mathbf{B}/B, \quad \mathbf{b} = \mathbf{T} \times \mathbf{n}$

$$M = \begin{bmatrix} a & q \\ p & d \end{bmatrix} = \frac{\theta}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} \delta_g & \beta_g \\ \beta_g & -\delta_g \end{bmatrix} + \frac{\alpha}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

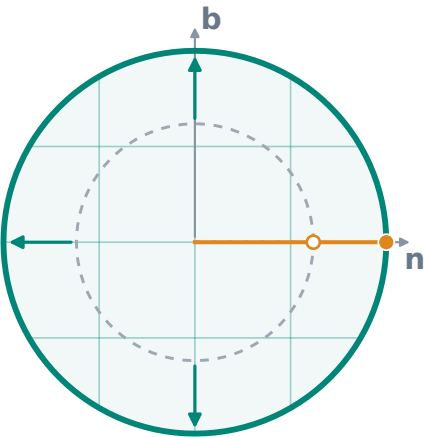
ISOTROPIC DILATION

Uniform expansion

$\theta > 0$



Normal-plane view



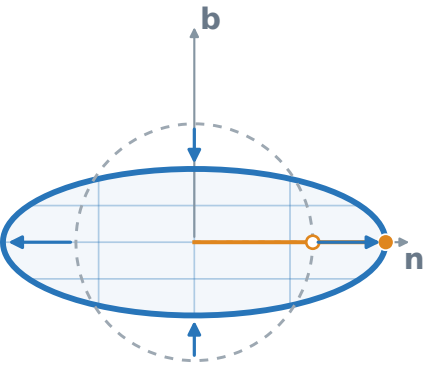
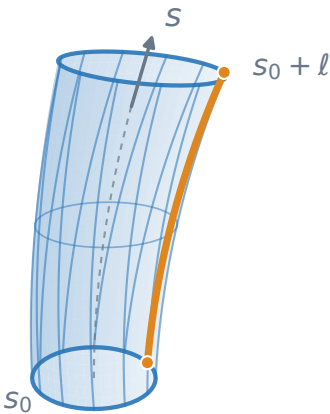
$\theta = a + d$

$A_1/A_0 = e^{\theta \ell}$

SYMMETRIC STRAIN / BIAxIAL SPLAY

Stretch along n; squeeze along b

$\delta_g > 0$

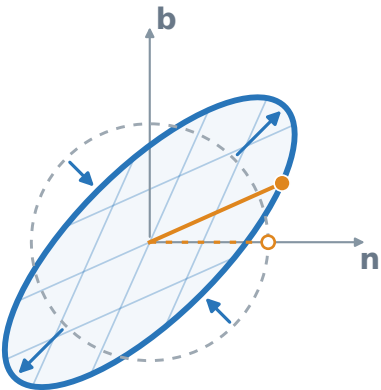
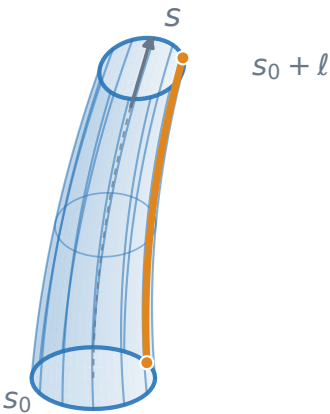


$\delta_g = a - d$

$A_1 = A_0$

Stretch and squeeze at 45 degrees

$\beta_g > 0$



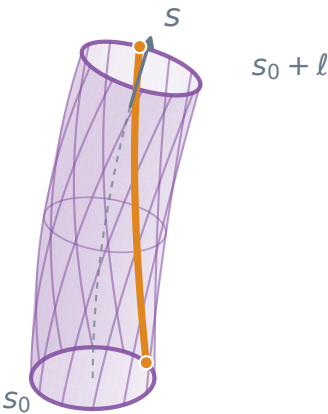
$\beta_g = p + q = D/B$

$A_1 = A_0$

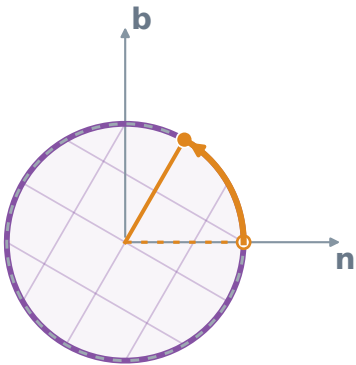
LOCAL ROTATION

Rotate marked field lines

$\alpha > 0$



n right, b up; T points towards the viewer



$\alpha = p - q = \mu_0 j_{\parallel} / B$

$\Delta\phi = \alpha \ell / 2, \quad A_1 = A_0$

Dashed grey: initial section Colour: mapped section Orange: the same labelled field line

Negative signs reverse the illustrated map.

$\xi(s_0 + \ell) = \exp(\ell M) \xi(s_0)$

Constant local coefficients; transported frame (torsion = 0 here). s is distance, not time.